

TACHOMETER WITH LED DISPLAY To Show Bicycle And Fan Speeds

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Measuring the speed of a rotating object is a common requirement. Knowing the speed of table and ceiling fans, as well as kitchen appliances such as mixers and grinders, can often be useful. However, fan speed controls only increase or decrease the speed and do not indicate the actual RPM (rotations per minute).

Cars, other vehicles, and even powered two-wheelers include a speedometer in front of the driver. This displays speed in kilometres per hour (km/hr), which is the commonly used unit. Earlier systems often expressed speed in miles per hour.

A tachometer device for bicycles can also be developed using an Arduino board based on the ATmega328P. Bicycle speedometers are useful for children who wish to monitor their speed while riding or racing.

The TM1637 four-digit LED display module is a compact and convenient device that shows bright numerical output using only a four-wire connection, as seen in the prototype. Fig. 1 shows the EFY Lab prototype (left) and the author's prototype used for measuring fan rotation (right).

For display, a TM1637-based LED segment display is used, and an IR proximity sensor is used for speed measurement. The components required to build this device are listed in the Bill of Materials table.

IR module, Arduino Uno, and TM1637 display

The IR module consists of an infrared LED and a photodiode that detect

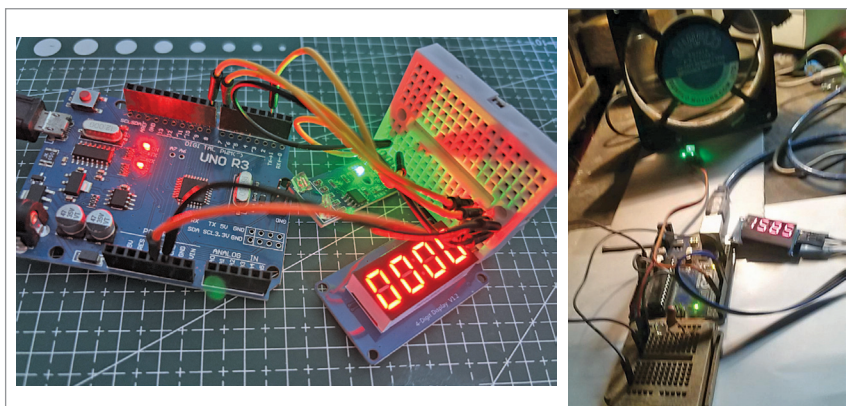


Fig. 1: EFY Lab prototype (left) and the author's prototype on fan rotation (right)

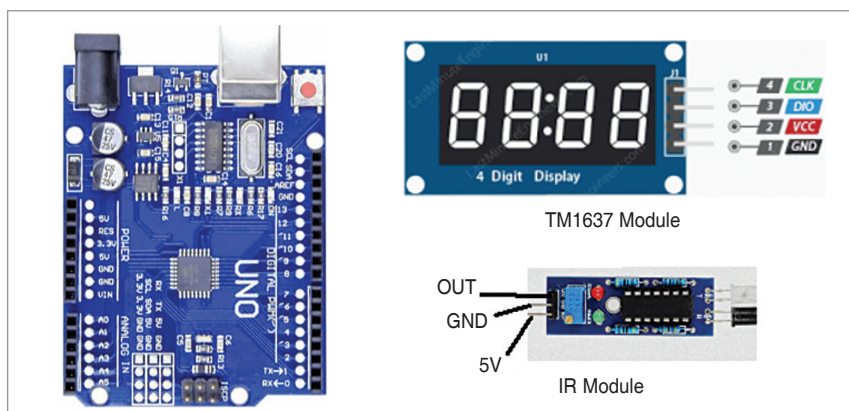


Fig. 2: Used pin details of Arduino, TM1637, and IR module

reflected infrared light from nearby objects and produce digital pulses. These pulses can be counted to detect motion, speed, or object presence.

The Arduino Uno is a microcontroller board based on the ATmega328P that processes signals from sensors and controls other devices.

The TM1637 4-digit 7-segment display module is a compact digital display module that shows numbers such as speed, time, or counts using only two communication pins.

Together, these components form a simple measurement system. Fig.

BILL OF MATERIALS	
Components	Quantity
Arduino Uno	1
TM1637-based LED segment display	1
IR sensor module	1
A small rod	1
Assembly item	As required
Jumper wires	As required

2 shows the Arduino Uno, TM1637 display, and the IR module used in this project.

The three pins of the IR module are Vcc (5V), ground, and output.